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Essay on Ethics in Engineering

Ethics are an essential part of being an engineer; as our level of accountability increases, our level of ethics must also increase. Some examples of the ethics engineers must embody are to accept responsibility in making decisions consistent with the safety, health, and welfare of the public, avoid conflicts of interest, be honest and realistic, reject bribery, accept criticism of technical work, and to treat all people fairly (IEEE). Many of the ethical violations are due to one or multiple of the “5 P’s”: pressure, pleasure, power, pride, and priorities (5 P’s PDH). When approaching a problem, analyze the 5 P’s and see where they could interfere with acting ethically in your current situation. Ensure that the decision you are making lines up with ethics. Picture someone else analyzing your work to see if it is ethical, you must hold yourself to that higher standard.

To prepare for this paper, I read about multiple examples of failed ethics. The example that stuck out the most was the infamous Ford Pinto disaster. During the development of the Pinto, prototypes were built and crashed, and the engineers determined that the fuel system in the rear of the car would fail during a rear-end collision. If there was an impact of 20 mph or more to the rear of the vehicle, the differential would dislodge, and an exposed flange and line of bolt heads would pierce the fuel tank and would cause a large explosion and fire directed towards the interior of the vehicle. This issue was ignored and covered up, because implementing an \$11 solution would be detrimental to Ford’s goal of vehicle cost being under \$2,000. This resulted in 180 burn deaths, 180 serious burn injuries, and 2100 burned vehicles (Pinto Issue PDH).

Looking back towards the “5 P’s”, there were multiple dilemmas where ethics were disregarded and contributed to this accident. There was pressure to produce the Pinto quickly, and to keep the vehicle under budget. The engineers were influenced by the power of their bosses, the engineers lied and signed off on the safety aspects of the car even though they were aware of the safety issues. The bosses were also held under the power of the higher expectations of the Ford Motor Company, possible shareholders, etc. The engineers were likely influenced by pride, there is humility needed to admit your design is flawed and needs fixed. Priorities were a large conflict in this case. The engineers and Ford prioritized their money over human lives. It was cheaper to pay out the settlements of burns, deaths, and vehicle loss than to pay to fix every Pinto with the new part to prevent this issue. You must always prioritize the safety of everyone involved.

Ethics needs to be presented at every level of production. It needs to be kept in mind in the early stages when brainstorming the solution, when creating the solution, testing the solution, producing the solution, marketing and selling the solution, and even post-sale ethics needs to be considered. Ethics also needs to be conserved in a vertical sense, the engineers need to keep ethics in mind, the supervisors and managers need to prioritize ethics, and the overall head of the business needs to promote and conserve a climate that is concerned and consistent with ethics.

Ethics are almost an ‘unspoken rule’. It is important to have strong ethics personally and be a moral person. These should be intrinsic traits and be ingrained in yourself and whoever you are working with. In a professional setting it is important that each person on the team is ethical and has integrity, and that a culture of ethics should be presented and promoted.

Works Cited

IEEE. *7.8 IEEE Code of Ethics*. PDF. Accessed Online 2 March 2024.

PDH Center. *5 P's*. PDH Online. PDF. Accessed Online 2 March 2024.

PDH Center. *Pinto Issue*. PDH Online. PDF. Accessed Online 2 March 2024.